# Louis-Roy Langevin

Montréal, QC, Canada
louis-roy.langevin@mail.mcgill.ca · louisroylangevin.github.io

## Education

2026–Present **University of Oxford** | *DPhil in Computer Science*
Supervisors: Prof. Paul Goldberg, Prof. Jiarui Gan

2024–2026 **McGill University** | *M.Sc. in Mathematics and Statistics (Thesis)*
Supervisor: Prof. Adrian Vetta
**Research Interests:** Algorithmic Game Theory, Combinatorial Optimization, Probability, Complexity Theory

2021–2024 **McGill University** | *B.Sc. Honours Mathematics and Computer Science*
Graduated with **First Class Honours and Distinction**
**CGPA:** 3.91 / 4.0

## Papers

2025 ***The Popular Dimension of Matchings***
with F. Connor, N. Ndiaye, A. Totschnig, R. Vasishta, & A. Vetta
Proceedings of 21st Conference on Web and Internet Economics (WINE), 2025.

2024 ***Optimal Root Recovery for Uniform Attachment Trees and $d$-regular Growing Trees***
with L. Addario-Berry, C. Fontaine, R. Khanfir, & S. Têtu
Accepted to Annals of Applied Probability.

2024 ***Optimally Revealing Bits for Rejection Sampling***
with A. Waese-Perlman
MIT Undergraduate Research Technology Conference, 2025.

2025 ***On Mobile Ad Hoc Networks for Coverage of Partially Observable Worlds***
with G. Dudek, A. Loria, E. Mériaux, S. Wen
Submitted to IEEE Transactions on Robotics (T-RO).

2025 ***Robotic Arm Rotation: Standing Up Is Harder Than You Think***
with N. Bousquet, F. Connor, R. El Sabeh, A. Mouawad, N. Nishimura, A. Totschnig
In 20th Scandinavian Symposium on Algorithm Theory (SWAT), 2026.

2025 ***Random Serial Dictatorship is $\sqrt{2}$-Envy-Free***
with F. Connor, M. Dupré la Tour, V. Narayan, N. Ndiaye, N. Rahman, A. Vetta
In 26th ACM Conference on Economics and Computation (EC), 2026.

2025 ***Tight Bounds on the Best $\theta$-Condorcet Winner of Metric Elections***
with F. Connor, N. Ndiaye, A. Totschnig, A. Vetta
To be submitted to Canadian Conference on Computational Geometry (CCCG), 2026.

## Research Internships

2024 **Root Recovery on Random Recursive Trees**
Worked under Prof. Louigi Addario-Berry on root-detection algorithms.

2023 **Burning Number Conjecture on Graphs**
Worked under Prof. Sergey Norin on cops and robbers and graph burning bounds.

2022 **Subelliptic Operators & Legendre Functions**
Studied linear combinations of Legendre functions (McGill / Dalhousie).

## Professional Experience

2025–Now **Mathematics Expert at *Meta Platforms, Inc.***
Spearheaded the analysis of advanced mathematics articles and designed/solved complex problems to enhance Large Language Model (LLM) advanced reasoning capabilities.

2024–2025 **Mathematics Specialist at *Scale AI, Inc.***
Designed/solved complex problems to enhance LLM advanced reasoning capabilities.

## Teaching & Mentoring

Fall 2025 **Substitute Lecturer** | McGill University
Taught graduate level lectures on algorithms and complexity theory (MATH553).

**Teaching Assistant** | McGill University

Fall 2025 – Algorithmic Game Theory

Winter 2025 – Calculus 2

Fall 2024 – Calculus 1

**Course Assistant** | McGill University

Winter 2023 – Discrete Mathematics

Fall 2022 – Honours Analysis 1

Winter 2025 **Research Mentor** | McGill Directed Reading Program
Supervised two undergraduate students during one semester for game theory research.

2024 + 2025 **Summer Camp Instructor** | McGill Programming Summer Camp
Taught algorithms and data structures to high school and college students.

2022-2025 **Founder & Educator** | *Cégepien* Online Platform
Created an online math preparation program with over 50 participants.
Produced educational videos with 17,000+ followers (@le_cegepien).

## Service & Leadership

2025 **Conference Paper Reviewer** | Symposium on Theoretical Aspects of Computer Science (STACS 2026).

2025 **Conference Co-Organizer**, 27th Institut des Sciences Mathématiques Graduate Student Colloquium.

**Conference Speaker:**

2025 – MIT Undergraduate Research Technology Conference (MIT URTC 2025).

2025 – Institut des Sciences Mathématiques Graduate Student Colloquium.

2025 – Seminars in Undergraduate Mathematics in Montreal (SUMM 2025).

2024 **Hackathon Judge**, McGill Code.jam(), 2nd place winner in 2023.

## Scholarships & Grants

2024–2026 **Graduate Excellence Award** | McGill University.

2025–2026 **Kenneth Eade Fellowship** | McGill University, awarded based on academic merit.

2025 **Lin Graduate Award** | McGill University.

2023 + 2024 **Undergraduate Student Research Award** | Natural Science and Engineering Research Council of Canada.

2022 + 2024 **Undergraduate Research Scholarship** | Institut des Sciences Mathématiques.

## Honors & Awards

2024 **First Class Honours with Distinction** | McGill University.

2024 **Third place at ICPC North America Northeast (NENA)** | McGill University.

2024 **Team Canada Representative, International Physics Tournament** | ETH Zürich.

2023 **W.L. Putnam Competition** — Top 600 in North America.

2020 **Canadian Open Mathematics Challenge** — 3rd National.

2019 **Association Mathématique du Québec Contest** — 1st National.